Job Market Analysis

DATS 6401 – Data Visualization

Dr. Anya Mendenhall

Josh Schechter
Dhanush Mathivanan
Halima Al Balushi

The George Washington University

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Background

In a previous analysis, we investigated the effects of significant economic downturns, identified as critical dates, on the overall health of the job market. In that analysis, the scope was more holistic, as we examined the job index as an aggregation of multiple sectors to provide a condensed and generalized view of job market performance. Figure 1 below best illustrates the findings from the previous study.

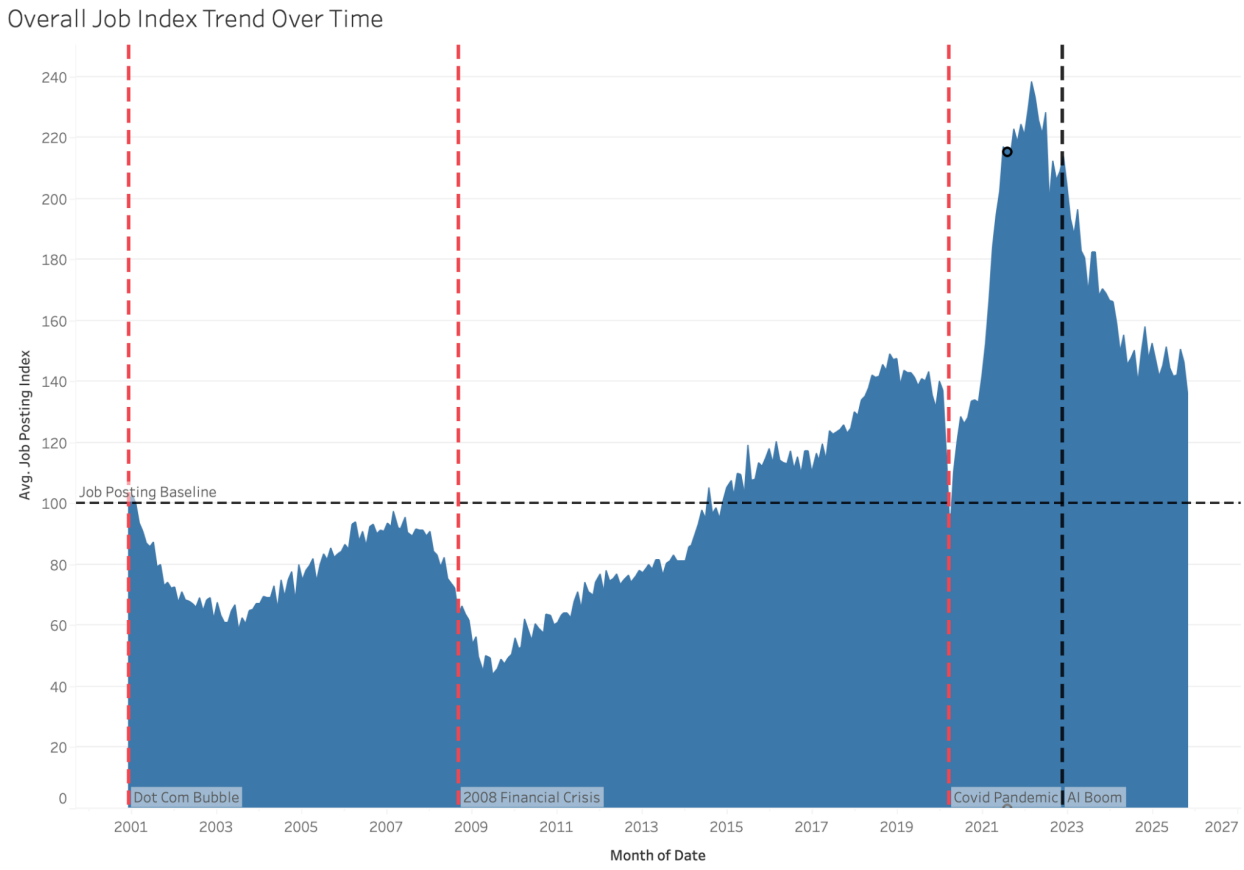

Figure 1: Aggregated Job Index Trend Over Time (From original study)

From this investigation, several observations were made. First, we examined the effects of standard economic downturns, such as the dot-com bubble and the 2008 financial crisis, both

identified as critical dates in Figure 1. In response to these events, job posting trends behaved similarly, showing a significant gradual decline followed by a slow, gradual recovery back to the baseline. Notably, it took a considerable amount of time for the job market to return to baseline levels.

The other two critical dates did not trigger the same response. COVID was especially interesting, as job postings experienced an immediate downturn, followed by an overcorrection, which then led to another gradual decline resulting in the current state of the job market. During this period of decline, the AI Boom occurred, which we defined as the release of OpenAI's product, ChatGPT. Our initial hypothesis was that the AI Boom was negatively affecting the overall health of job openings; however, this could not be conclusively determined from this investigation alone.

This study builds on those findings with the goal of answering the following questions:

1. Does the year-to-year change in the job index align with changes in population?
2. When evaluating job openings at the sector level, do the trends match those observed in the aggregated view?
3. Does unemployment increase in states that experience larger year-to-year population changes? If so, what are some potential explanations?

Datasets

For our initial project analysis, we used five datasets from two primary sources, with the addition of a population field to improve the accuracy of state-level comparisons.

The initial dataset was sourced from Kaggle and includes unemployment metrics by U.S. state (*Unemployment in USA*):

https://www.kaggle.com/datasets/justin2028/unemployment-in-america-per-us-state

While the dataset appears to have been created for independent statistical analysis, it provides reliable state-level unemployment data used in our study.

The remaining datasets were obtained from the Federal Reserve Economic Data (FRED) platform: https://fred.stlouisfed.org/

These datasets provide job opening rates across multiple sectors, along with state-level population data to support a more accurate assessment of labor market trends.

For the final project, we extended our analysis by incorporating **state-level population estimates for the United States from 2000 to 2025**:

https://fred.stlouisfed.org/series/CAPOP - Selection based on view map option (US States)

To analyze job market trends, we utilized sector-specific job openings datasets from the Job Openings and Labor Turnover Survey (JOLTS), including:

- Information Technology job postings:
  https://fred.stlouisfed.org/series/JTU5100JOL
- Professional & Business Services job postings:
  https://fred.stlouisfed.org/series/JTS540099JOL
- Finance & Insurance job postings:
  https://fred.stlouisfed.org/series/JTU5200JOL
- Total Non-Farm job postings:
  https://fred.stlouisfed.org/series/JTSJOL

These datasets provide time-series data across different industries, enabling a more granular comparison of job market behavior across sectors.

All datasets were cleaned, standardized, and aligned using common attributes such as time (year/month) and geographic identifiers (state/region). This enabled the integration of population, job openings, and unemployment data into a unified analytical framework for comparative analysis.

While these datasets provide strong coverage of macro-level trends, they remain aggregated at the sector and state level. As a result, the analysis does not capture variations at the individual job role or company level.

Data Story

To address the research questions outlined in the background section, we derived two new calculated fields within Tableau Desktop to measure change. These derived fields are referred to throughout the report as **Average Job Posting Percent Change** and **Average Population Percent Change**. Additionally, we used our two primary metrics from the initial paper, as we were able to collect more detailed sector-related data. These metrics include unemployment by state and the average job posting index.

Our reference lines remain consistent, with a baseline set at the start date and four critical dates identified as the dot-com bubble, the 2008 financial crisis, COVID, and the AI Boom.

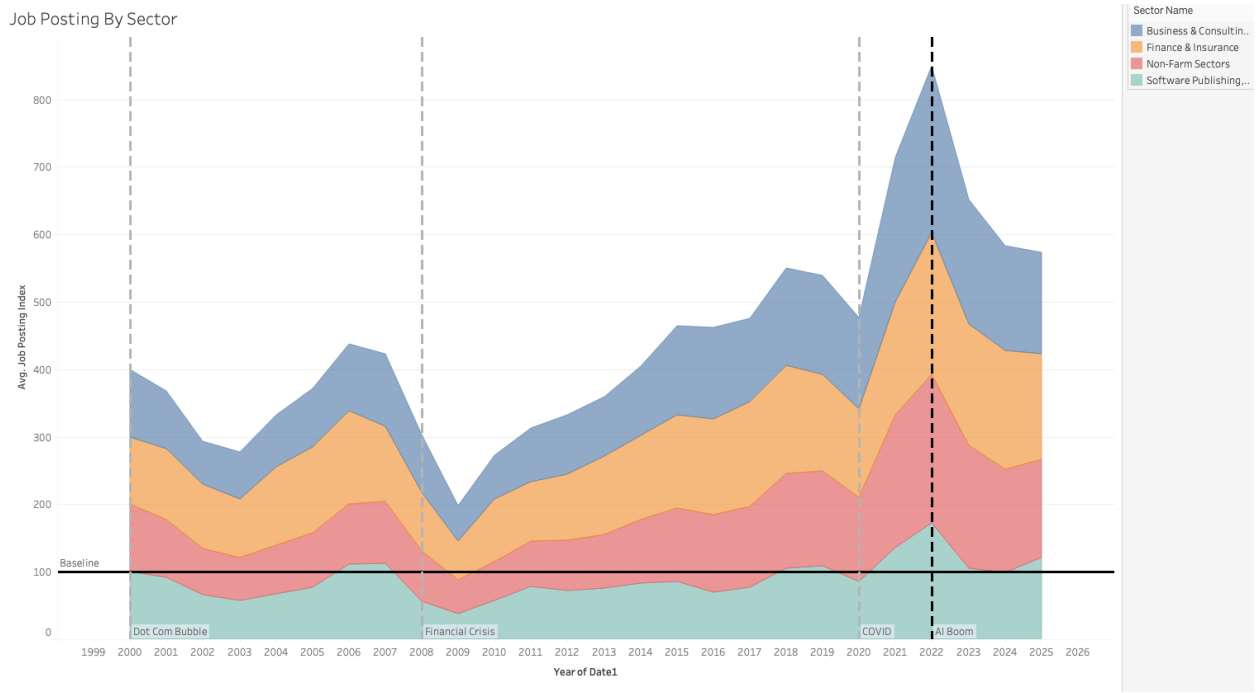

Figure 2: Job Posting Index by Sector

Figure 2 can be viewed as an extension of Figure 1, which was introduced in the initial paper. The key difference is that Figure 2 presents a time series of job index changes by sector, rather than an aggregation across all sectors. Each color represents a different sector, with the red area corresponding to the most general job category. Specifically, general business is shown in blue, finance in orange, and software in green.

Interestingly, all sectors follow a similar overall trend. They decline during economic downturns and increase during periods of economic growth. This suggests that the aggregated view shown in Figure 1 is sufficient for capturing general job market trends over time. However, this visualization does not demonstrate whether job index growth aligns with population changes, indicating that further analysis is needed.

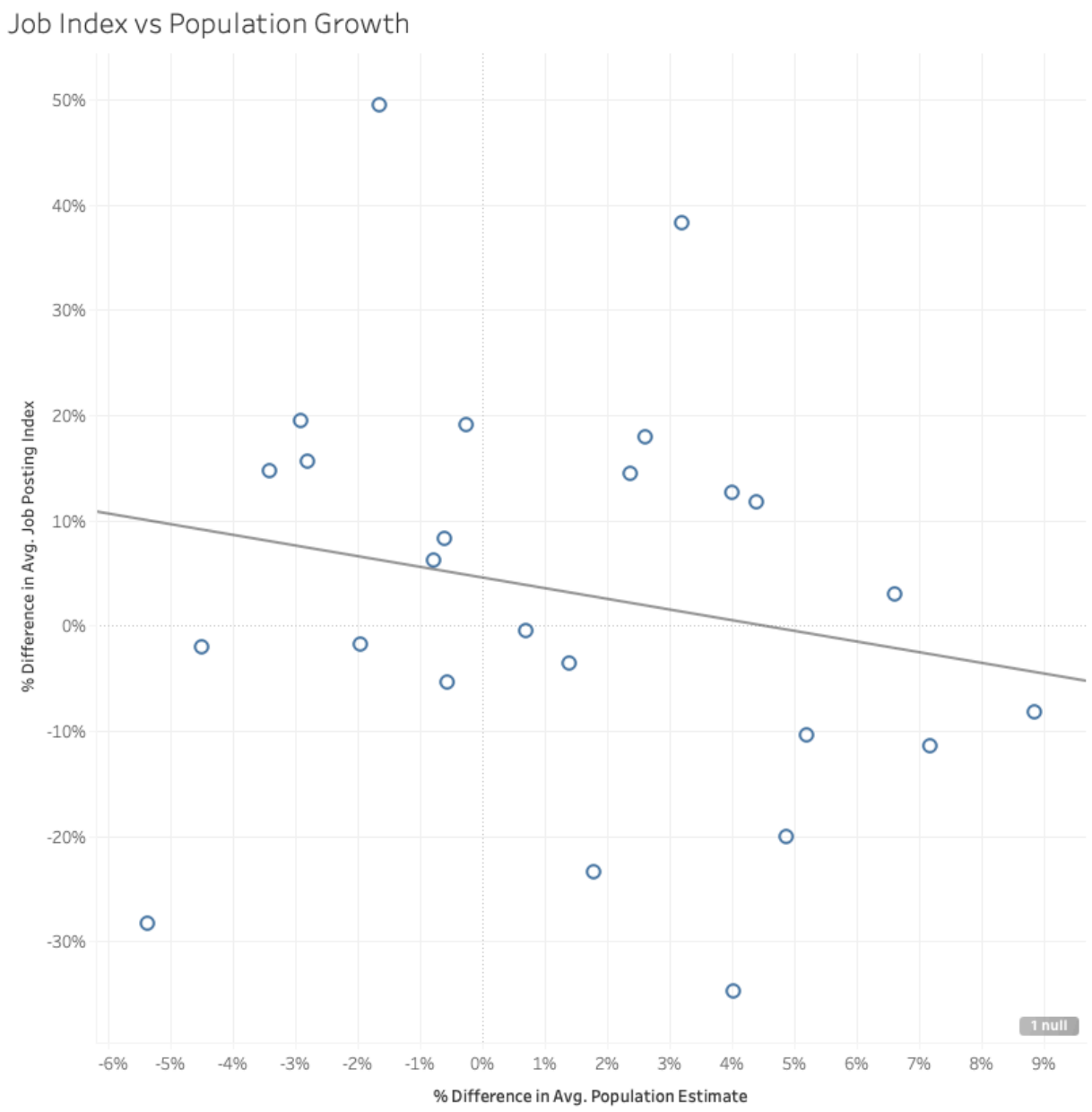

Figure 3: Aggregated Job Index Growth vs Population Growth

Figure 3 uses our newly derived metrics to examine whether there is a correlation between population growth and job growth. On the x-axis, we show the year-to-year percent change in population, and on the y-axis, we show the percent change in the average job posting index.

For the job market to be considered healthy, we would expect to observe a positive, direct correlation between these two metrics. This would indicate that population-driven demand for jobs is being met by the job market. However, the results suggest the opposite.

Across the entire time period analyzed, there appears to be a negative correlation between population growth and job index growth. Notably, this is based on an aggregated view, and Figure 4 below further breaks this relationship down at the sector level.

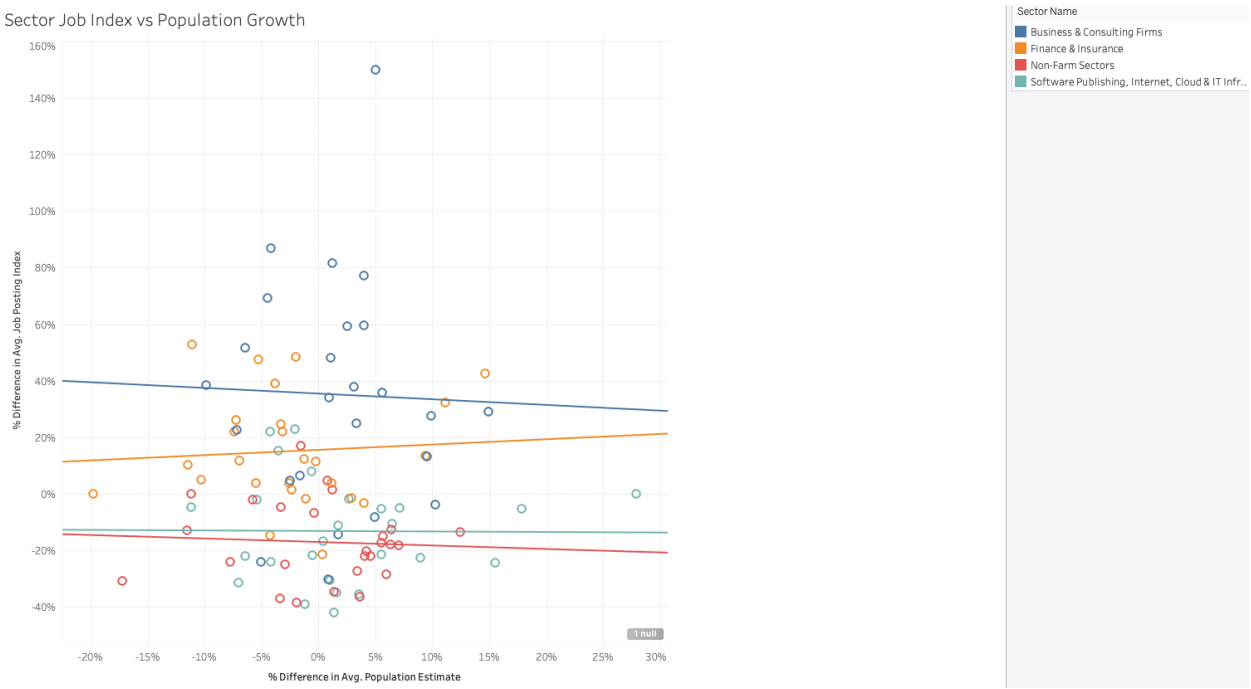

Figure 4: Sector Based Job Index Growth vs Population Growth

Figure 4 presents a sector-level view of the relationship shown in the previous figure. The colors of the four sectors are consistent with Figure 2: green represents software, orange represents finance, blue represents business, and red represents general sectors.

Interestingly, not all sectors behave in the same way. As indicated by the regression lines, finance job postings show a slight positive correlation with population growth, while the software sector effectively has no correlation.

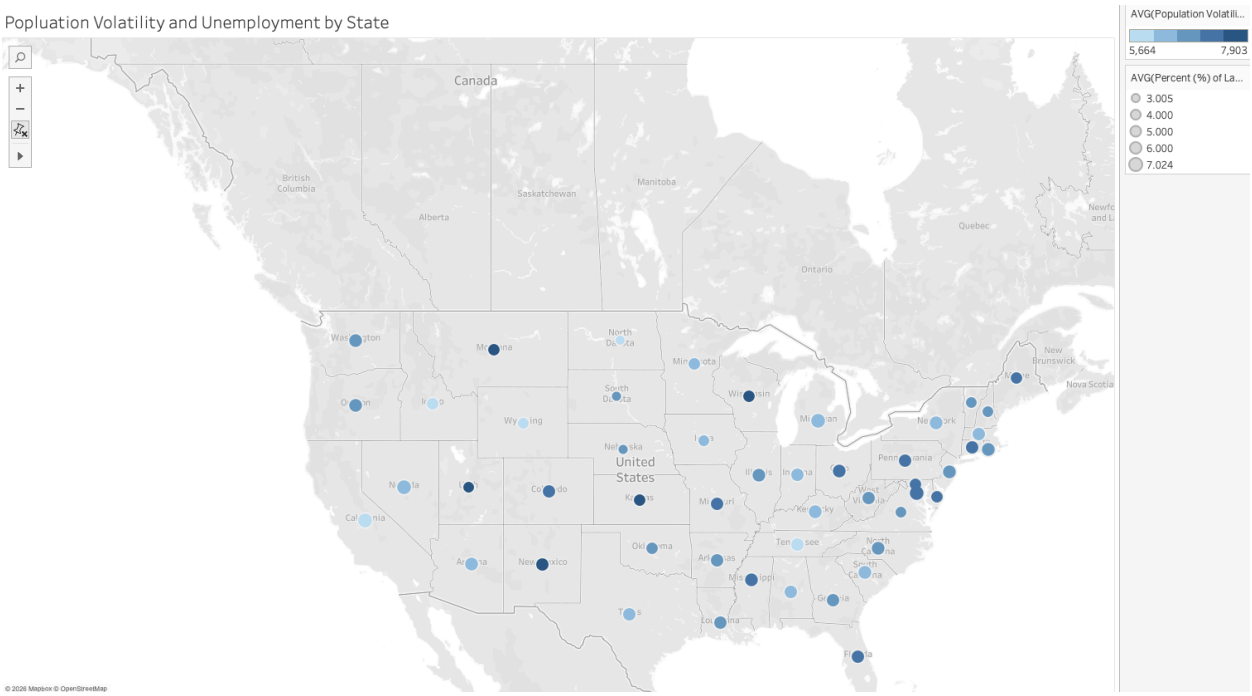

Figure 5: Population Volatility vs Unemployment by State

Figure 5 aims to address Research Question 3 by examining whether states with higher population volatility experience higher average unemployment rates.

Overall, this visualization does not reveal a clear or consistent relationship between population volatility and unemployment. As a result, it is difficult to draw definitive conclusions from this analysis alone.

Analysis and Conclusion

This project expanded on Project 1 in two meaningful ways. First, it extended the analysis from an aggregated job index view to a sector-based job index view, allowing for a more robust examination of specific sectors. Ultimately, the results were similar, as the trends observed within individual sectors closely matched those of the aggregated view.

Second, this analysis accounted for population variance when evaluating job index and unemployment changes. Instead of examining job index change alone, we considered population change alongside it. This approach allowed for a more accurate assessment of the overall health of the job market. For example, if we consider Figure 1 in isolation, without accounting for population, we might conclude that the job market has returned to its pre-COVID 2019 level. However, when accounting for population growth, it becomes clear that jobs per capita have declined, as the job index has not grown or stabilized sufficiently to meet the needs of the increasing population.

Moreover, advanced Tableau features, such as the creation of calculated fields and the use of dashboard actions, enabled a more robust analysis of change over time. In time series analysis, examining percent change is essential for understanding underlying relationships.

References

Oh, J. (2026). *Unemployment in America, per U.S. state* [Dataset]. Kaggle. https://doi.org/10.34740/KAGGLE/DSV/14503856

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